Activity 8.2: Transformations

A

Name: _____ Partner's name: _____ Date: _____

Directions: Find the coordinates of the image, or describe the transformation. All rotations and dilations are about the origin.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Translate $(2, 3)$ using the rule $(x, y) \to (x+4,\ y-5)$.

Answer: _____ Match? ☐ Yes ☐ No

2

Reflect $(5, -3)$ over the x-axis.

Answer: _____ Match? ☐ Yes ☐ No

3

Reflect $(2, 7)$ over the line $y = x$.

Answer: _____ Match? ☐ Yes ☐ No

4

Rotate $(3, 4)$ $90°$ counterclockwise.

Answer: _____ Match? ☐ Yes ☐ No

5

Rotate $(-3, 5)$ $180°$.

Answer: _____ Match? ☐ Yes ☐ No

6

Dilate $(3, -2)$ by a scale factor of 2.

Answer: _____ Match? ☐ Yes ☐ No

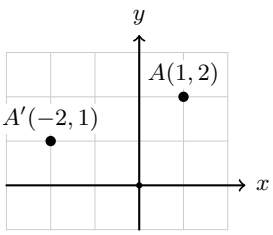
7

Translate $(1, 3)$ using $(x, y) \to (x + 2,\ y + 1)$, then reflect the result over the y-axis.

Answer: _____ Match? ☐ Yes ☐ No

8

$A(1, 2)$ is mapped to $A'(-2, 1)$ by a rotation. Describe the rotation.



Answer: _____ Match? ☐ Yes ☐ No

Activity 8.2: Transformations

B

Name: _____ Partner's name: _____ Date: _____

Directions: Find the coordinates of the image, or describe the transformation. All rotations and dilations are about the origin.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Translate $(-1, 6)$ using the rule $(x, y) \to (x+7,\ y-8)$.

Answer: _____ Match? ☐ Yes ☐ No

2

Reflect $(-5, 3)$ over the y-axis.

Answer: _____ Match? ☐ Yes ☐ No

3

Reflect $(-2, -7)$ over the line $y = -x$.

Answer: _____ Match? ☐ Yes ☐ No

4

Rotate $(-3, -4)$ $90°$ clockwise.

Answer: _____ Match? ☐ Yes ☐ No

5

Reflect $(-3, 5)$ over the x-axis, then reflect the result over the y-axis.

Answer: _____ Match? ☐ Yes ☐ No

6

Dilate $(12, -8)$ by a scale factor of $\frac{1}{2}$.

Answer: _____ Match? ☐ Yes ☐ No

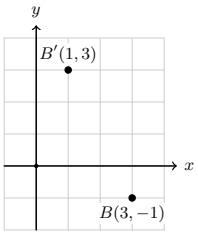
7

Reflect $(5, 3)$ over the y-axis, then translate the result using $(x, y) \to (x+2,\ y+1)$.

Answer: _____ Match? ☐ Yes ☐ No

8

$B(3, -1)$ is mapped to $B'(1, 3)$ by a rotation. Describe the rotation.



Answer: _____ Match? ☐ Yes ☐ No